

Mesenteric Pathologic Conditions: Interactive Case-based Approach

Amelia L. Kernizan, MD • Jonathan Revels, DO • Cristina Hajdu, MD • Maria Manning, MD • Myles T. Taffel, MD

Author affiliations, funding, and conflicts of interest are listed at the [end of this article](#).
See the slide presentation [here](#).

The abdominal mesentery consists of a double-layered peritoneal reflection that suspends the bowel from the posterior abdominal wall and encases the mesenteric vasculature, the lymphatics, and a variable amount of fat. The differential diagnosis for mesenteric pathologic conditions is broad, ranging from benign and purely incidental causes, such as mesenteric panniculitis and retractile mesenteritis, to malignant and aggressive causes, such as neuroendocrine tumors, lymphoma, and metastatic disease.

Overlapping imaging features of mesenteric pathologic conditions present a diagnostic dilemma for radiologists. When present, clinical history and symptoms can aid in differentiation. However, many mesenteric pathologic conditions are often discovered incidentally. Both factors contribute to the frustration felt by many radiologists when faced with determining the potential diagnosis and/or providing the most appropriate recommendations for further assessment of these lesions.

Sclerosing mesenteritis is an umbrella term used by pathologists to define nonspecific inflammatory changes of the mesentery; the disease spectrum varies from mesenteric panniculitis to retractile mesenteritis. The histologic features underscore the radiologic findings in that the lipid-laden macrophages and areas of fat necrosis seen in mesenteric panniculitis correlate with hazy fat stranding with a thin pseudocapsule and small mesenteric lymph nodes that typically measure less than 1 cm. Patients with sclerosing mesenteritis are typically asymptomatic. The presence of abdominal symptoms warrants close inspection for acute processes, such as enteritis or trauma.

Retractile mesenteritis is at the other end of the spectrum, with pathologic features of dense masslike fibrosis and inflammation. At imaging, retractile mesenteritis can appear masslike with calcifications but without lymphadenopathy. When lymphadenopathy is present, alternative diagnoses to sclerosing mesenteritis should be considered, such as *Mycobacterium avium* complex infection or neoplastic processes (ie, lymphoma).

While many mesenteric pathologic conditions will ultimately require tissue sampling for definitive diagnosis, there occasionally are clues gleaned through the patient's history and the radiologic features of the lesion that can help narrow the differential diagnosis. The slide presentation uses a quiz-style case-based approach to provide radiologists with a

framework for approaching solid mesenteric pathologic conditions. Additionally, we provide an algorithm that learners can use in their evaluation of mesenteric lesions that would lead to a favored diagnosis based on the patient's history of malignancy, presence of mesenteric stranding (Fig 1) or a soft-tissue mass (Fig 2), presence or absence of enlarged lymph nodes, and associated symptoms. The series of cases comprise clinical, radiologic, and pathologic correlations across a wide variety of commonly encountered mesenteric abnormalities, in addition to some of the less commonly encountered entities so that radiologists will have a familiarity with these as well. When possible, we provide illustrative examples of the underlying histologic features for select pathologic conditions and discuss how the underlying tissue structure influences the appearances demonstrated at imaging.

The slide presentation will help radiologists better understand the mesentery and its more common pathologic conditions as well as how to make more informed differential diagnoses and follow-up recommendations when these conditions are encountered at imaging.

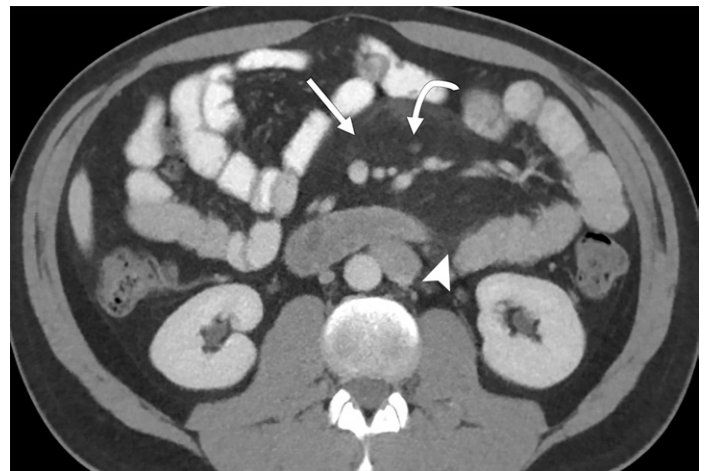


Figure 1. Mesenteric panniculitis in an asymptomatic 40-year-old man. Axial CT image shows incidentally found nonspecific haziness of the mesentery (straight arrow) with a thin pseudocapsule (arrowhead) and small mesenteric lymph nodes measuring less than 1 cm (curved arrow). No associated soft-tissue mass or retroperitoneal lymph nodes are present.



RadioGraphics 2023; 43(12):e230077
<https://doi.org/10.1148/rg.230077>
 Content Codes: CT, GI, MR

TEACHING POINTS

- Sclerosing mesenteritis can be pathologically divided into two subgroups that correlate with imaging findings. Mesenteric panniculitis manifests with mesenteric haziness with small lymph nodes, while retractorile mesenteritis manifests with a focal mesenteric mass with calcifications.
- Metastatic carcinoid tumor can manifest similarly to retractorile mesenteritis with a calcified mesenteric mass and fibrosis. Somatostatin imaging or tissue sampling is frequently needed to differentiate these entities.
- Enlarged mesenteric lymph nodes may be seen in both benign and malignant conditions, but a short-axis diameter greater than 1 cm is more concerning for malignancy and should prompt further workup or tissue sampling.

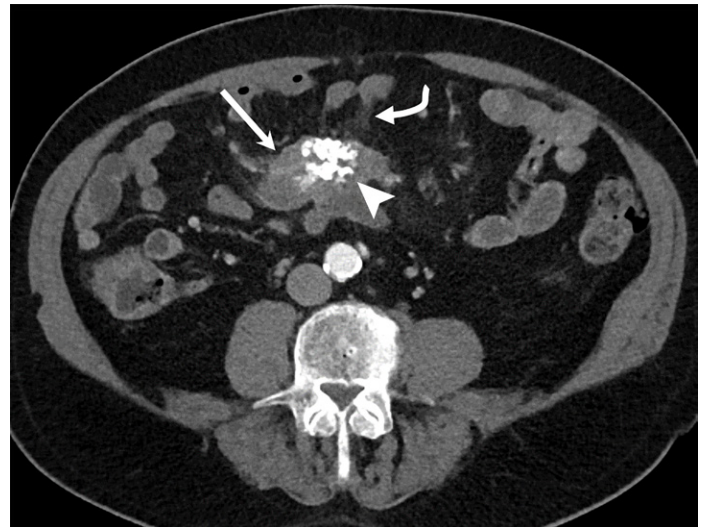


Figure 2. Retractorile mesenteritis in a 78-year-old man who presented with abdominal pain. Axial CT image shows an incidentally found mesenteric soft-tissue mass (straight arrow) with heterogeneous central calcifications (arrowhead). Multiple adjacent small bowel loops are tethered to the mass (curved arrow) without associated bowel obstruction. The mass was found to represent retractorile mesenteritis.

Author affiliations.—From the Department of Radiology, NYU Grossman School of Medicine, 660 1st Ave, New York, NY 10016 (A.L.K., J.R., C.H., M.T.T.); and ACR Institute for Radiologic Pathology, Silver Spring, Md (M.M.). Presented as an education exhibit at the 2022 RSNA Annual Meeting. Received April 4, 2023; revision requested May 25 and received June 22; accepted June 28. **Address correspondence to** A.L.K. (email: Amelia.Kernizan@nyulangone.org).

Disclosures of conflicts of interest.—M.M. Royalties from Elsevier; payment for medical advice provided for possible medical malpractice case; travel support from MedStar Georgetown University Hospital and the American College of Radiology Institute for Radiologic Pathology; *RadioGraphics* contributing editor; gastrointestinal section chief for the American College of Radiology Institute for Radiologic Pathology. M.T.T. Payment for expert review from German, Gallagher & Murtagh. All other authors have disclosed no relevant relationships.

Suggested Readings

Burrill J, Williams CJ, Bain G, Conder G, Hine AL, Misra RR. Tuberculosis: a radiologic review. *RadioGraphics* 2007;27(5):1255–1273.
 Camacho JC, Moreno CC, Harri PA, Aguirre DA, Torres WE, Mittal PK. Post-transplantation lymphoproliferative disease: proposed imaging classification. *RadioGraphics* 2014;34(7):2025–2038.
 Corwin MT, Smith AJ, Karam AR, Sheiman RG. Incidentally detected misty mesentery on CT: risk of malignancy correlates with mesenteric lymph node size. *J Comput Assist Tomogr* 2012;36(1):26–29.

Gezer NS, Başara I, Altay C, et al. Abdominal sarcoidosis: cross-sectional imaging findings. *Diagn Interv Radiol* 2015;21(2):111–117.
 Hubbard R, Kotecha J, Nash T, Lee YJ, Khan N, Kazmi F. Radiology examination as a diagnostic aid in presentations with wide differential diagnoses: Case report of new Hodgkin's lymphoma on a background of poorly controlled HIV. *SA J Radiol* 2017;21(2):1239.
 Ruuska T, Ramírez Escalante Y, Vaittinen S, et al. Somatostatin receptor expression in lymphomas: a source of false diagnosis of neuroendocrine tumor at 68Ga-DOTANOC PET/CT imaging. *Acta Oncol* 2018;57(2):283–289.
 Sato M, Hiyama T, Kaito K, Hayashi Y, Okumura T. Usefulness of F-18 FDG PET/CT in the assessment of disseminated *Mycobacterium avium* complex infection. *Ann Nucl Med* 2009;23(8):757–762.
 Shinagare AB, Ramaiya NH, Jagannathan JP, et al. A to Z of desmoid tumors. *AJR Am J Roentgenol* 2011;197(6):W1008–W1014.
 Taffel MT, Khatri NJ, Hai N, Yaghmai V, Nikolaidis P. De-misty-fying the mesentery: an algorithmic approach to neoplastic and non-neoplastic mesenteric abnormalities. *Abdom Imaging* 2014;39(4):892–907.
 Yu RS, Zhang WM, Liu YQ. CT diagnosis of 52 patients with lymphoma in abdominal lymph nodes. *World J Gastroenterol* 2006;12(48):7869–7873.